Human Systems Integration (HSI)

Key Concepts & NASA HSI Plan Guidance

Bethany Ryan
University of Idaho
PSYC 562: Advanced Human Factors
Professor Roger Lew
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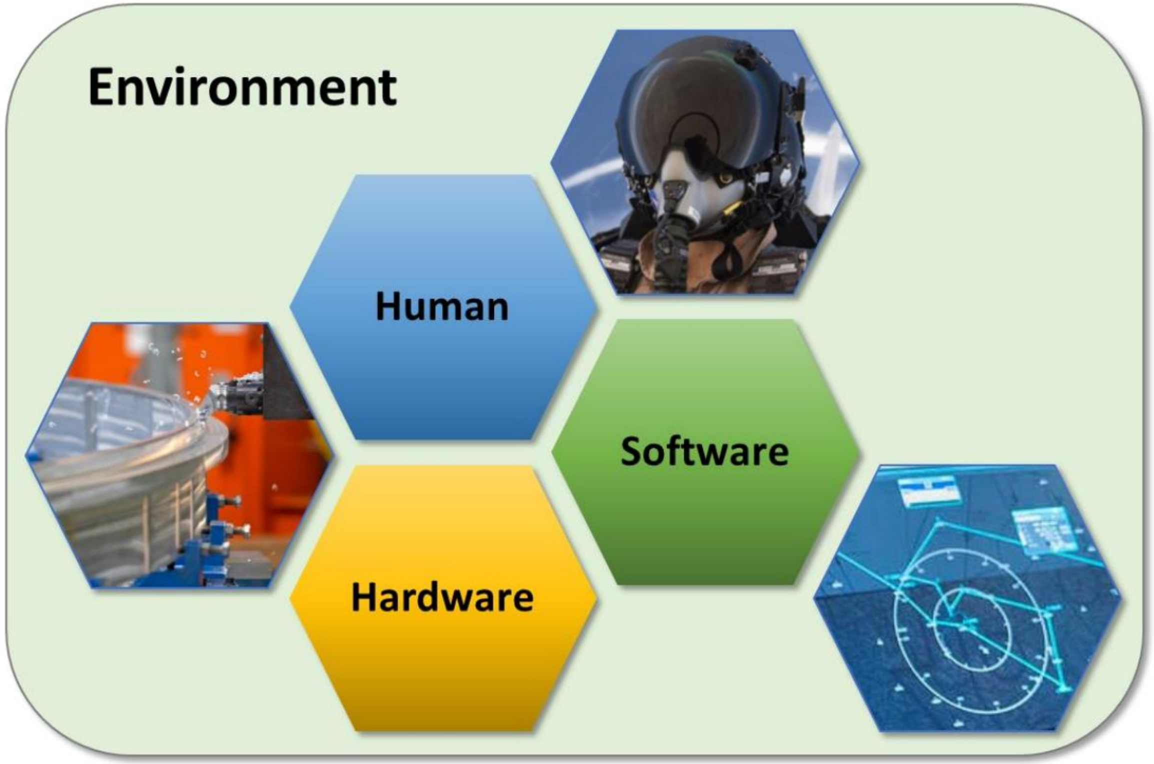


Figure 1-1. HSI System: Integrated Hardware, Software, and Human Elements within an Environment

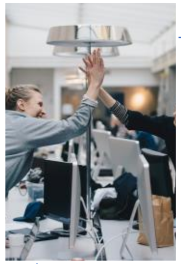
What is Human Systems Integration?

As defined by the NASA HSI Community of Practice, HSI is a required interdisciplinary integration of the human as an element of a system to ensure that the human and software/hardware components cooperate, coordinate, and communicate effectively to successfully perform a specific function or mission.

Key HSI Concepts

System Composition

Human Interactions



Integration and Collaboration

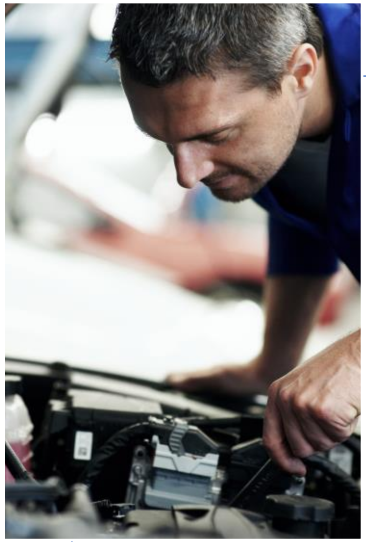
Early and Iterative Application

Key HSI Concepts

System Composition

- HSI integrates hardware, software, and human elements.
- Human element is critical for overall system performance and efficiency.
- Systems engineering approach applies to all elements throughout the project lifecycle.

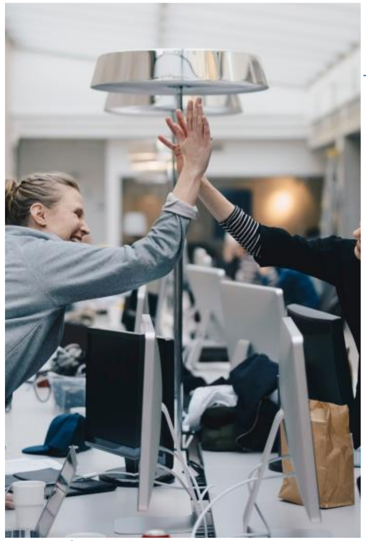
Key HSI Concepts



Human Interactions

- Consider all personnel interfacing with the system at any lifecycle phase.
- Includes end users (pilots, crewmembers), maintainers, ground controllers, logistics personnel, sustaining engineers, etc.

Key HSI Concepts



Integration and Collaboration

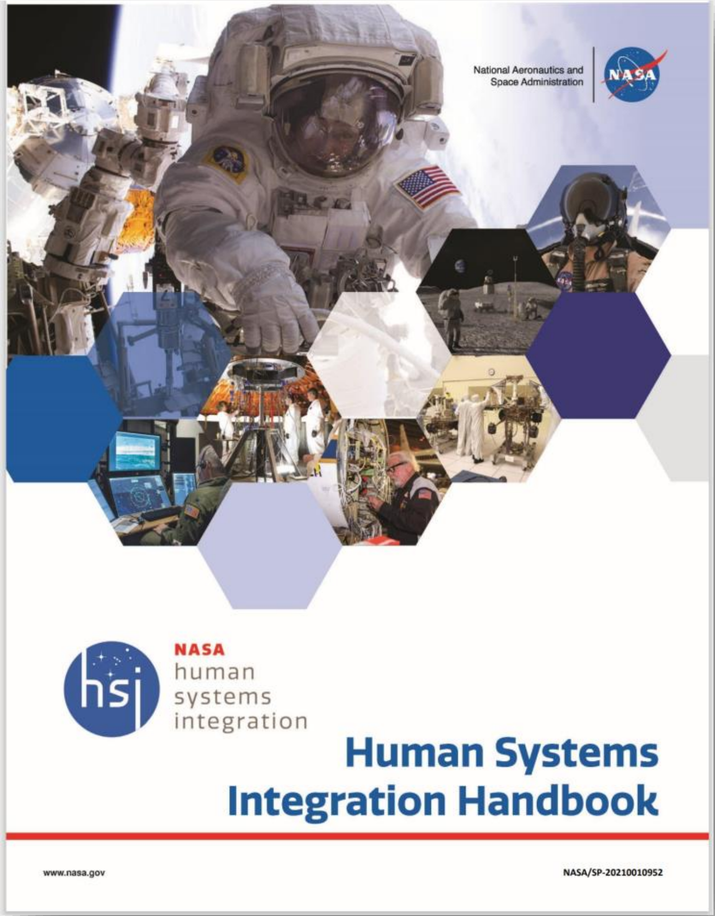
- Successful HSI requires collaboration across multiple domains.
- HSI ensures coordinated human-centered input in system design and engineering.
- HSI team leads resolve conflicts among human systems experts before involving P/P management.

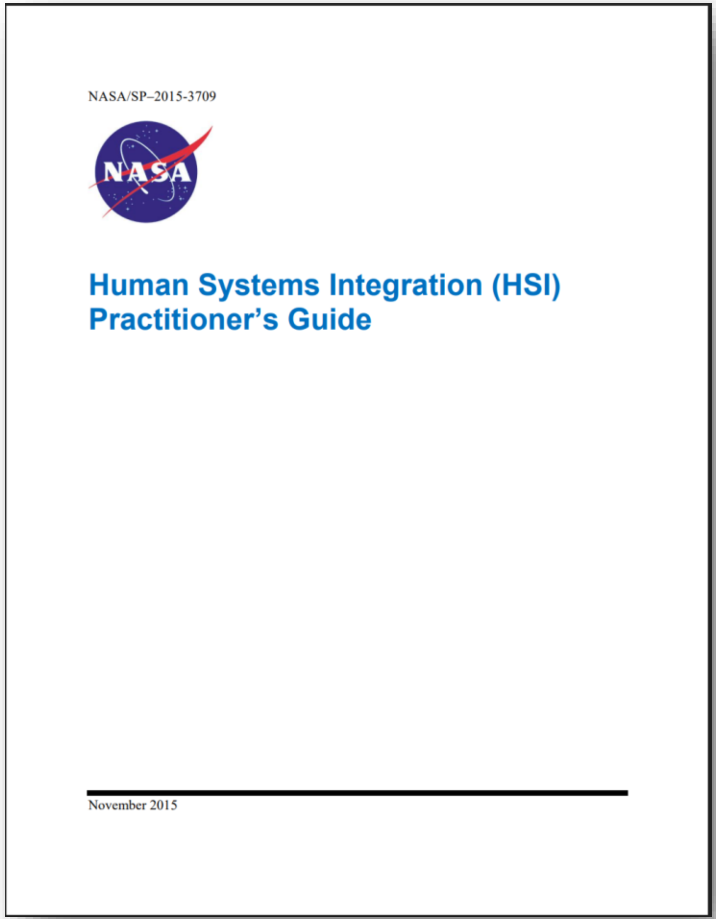
Key HSI Concepts

Early and Iterative Application

- HSI must be integrated early in system development and acquisition.
- Applied iteratively from early concept and design phases through to final phases.
- Early HSI application maximizes lifecycle cost efficiency and system performance.
- HSI requirements drive metrics and embed goals within system design.

NASA Guidance




NASA/SP–2015-3709

Human Systems Integration (HSI) Practitioner's Guide

November 2015

NASA HSI Plan Template

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HSI Plan – 1.0: Introduction & 2.0: Relevant Documents

Purpose

- Defines processes and roles for HSI activities.
- Documents trade-offs affecting human elements (operators, maintainers, etc.).
- Integrates human concerns with project objectives.

Scope

- Bridges communication between project management and HSI teams.
- Uses documentation, prototypes, and design reviews for compatibility.
- Tracks HSI efforts, risks, and decisions.

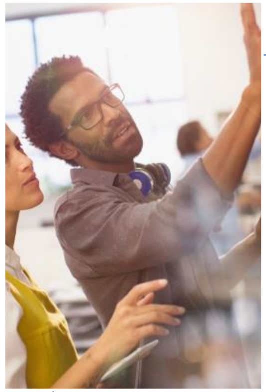
Applicable Documents

- Documents directly invoked by HSI efforts.

Reference Documents

- Supplemental information for HSI application.

HSI Plan – 3.0: HSI Strategy



HSI Program Roles and Responsibilities

- HSI Lead and Team: plan, manage, execute, and evaluate HSI activities.
- Collaborate with program management and SE teams.
- Develop and manage HSI requirements, ensure integration into lifecycle documents.
- Participate in design reviews, test planning, and risk management.
- Maintain HSI documentation and track activities.

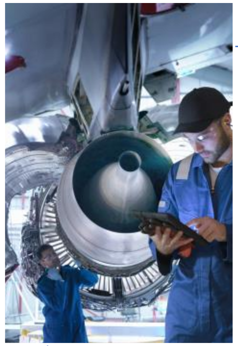
HSI Team Organization

- Define roles and responsibilities within the HSI team.
- Interaction with program management, SE teams, hardware/software teams, and technical authorities.
- Integrated Product Teams (IPTs) and Working Groups (WGs) for collaboration.
- Ensure HSI considerations are incorporated in system/sub-system design and testing.

HSI Plan – 4.0 HSI Domains

Human Factors Engineering

- Focuses on analysis, design, development, and testing of system interfaces.
- Ensures human performance characteristics are integrated into system design.
- Addresses workload, ergonomics, human-machine teaming, and safety.



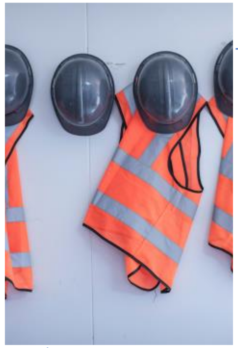
Operations

- Designs systems for robust, cost-effective operations.
- Leverages lessons learned from past programs.
- Considers human performance in procedure development and functional allocation..

Maintainability and Supportability

- Simplifies maintenance and optimizes resources.
- Addresses reliability, maintainability, and safety.
- Evaluates ease of installation, hazard elimination, and resource requirements.

HSI Plan – 4.0 HSI Domains



Habitability and Environment

- Considers impacts on human morale, safety, health, and performance.
- Includes environmental health, radiation, toxicology, nutrition, acoustics, and more.
- Ensures adequate lighting, noise control, and safety measures.

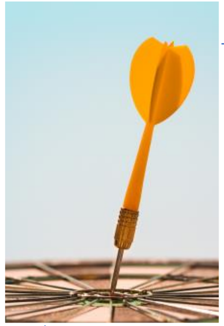
Safety

- Focuses on minimizing risks to personnel and mission.
- Includes safety analysis, human error analysis, reliability, and hazard controls.
- Ensures safe operation, maintenance, and emergency procedures.

Training

- Designs effective training programs to maximize human proficiency.
- Considers ease of training, specialized skills, and training methods.
- Ensures operators and maintainers are adequately trained for their tasks.

HSI Plan – 5.0: HSI Implementation



HSI Project Focus Areas

- Identifies key focus areas mapped to HSI domains.
- Monitors and addresses HSI issues to prevent program risks.
- Translates focus areas into domain-specific HSI planning actions.

HSI Issue and Risk Processing

- Identifies and mitigates human system risks.
- Ensures risks are documented and managed appropriately.
- Coordinates with project risk management to elevate HSI risks.

HSI Activities and Products

- Contributes to mission success, affordability, operational effectiveness, and safety.
- Integrates human performance characteristics into system design.
- Includes analyses, evaluations, HITL testing, simulations, and design reviews

HSI Plan – 6.0: Documentation of Lessons Learned

Documentation of Lessons Learned

- Critical for both new systems and iterations of existing systems.
- Ensures current experiences and solutions are included in future systems.
- Should be documented continuously, not just at the end of the program.

Human Systems Integration Introduction

YouTube · NASA Video · Apr 7, 2016